micro:bit

Introduction to Computer Science

Unit 9: Bits, bytes, and binary

Microsoft

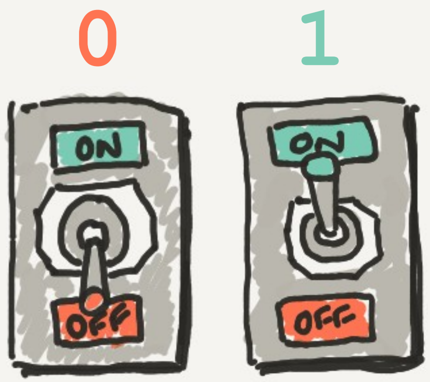
Lesson A

Understanding bits, bytes, binary

What we'll learn in Lesson A

- What bits and bytes are
- The binary number system

Bits, bytes, and binary



Bit	Byte							
0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1

Binary

- A numbering system based on **powers of two**

Bit

- The **smallest unit of information**
- Can **only be a zero or one**
- Represents the **state of one switch**
  - If the switch is **off**, then the bit has the value **zero**
  - If the switch is **on**, then the bit has the value **one**

Byte

- A sequence of **eight bits** that represents larger pieces of information
- Can represent any value **from 00000000 through 11111111**
- **Each digit in a byte** can be thought of as representing an **individual switch** that is either off (0) or on (1)

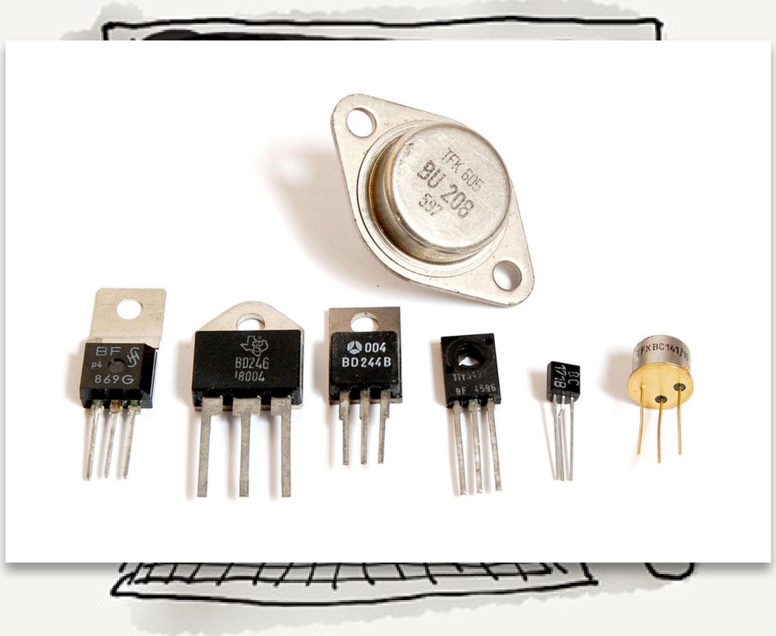
Packed transistors

Modern computer processors are made up of billions of tiny On/Off switches called **transistors**, but **information** is still represented in essentially the same way: **as a series of 1s and 0s**.

Using binary allows computers to represent information simply and efficiently.

In coding:

- The 1s and 0s of bits and bytes can be used to represent **letters**, **numbers**, and even different **keys** on a computer keyboard.
- A bit can be used to hold a **Boolean** (true/false) value. A value of **0 represents "false"** and a value of **1 represents "true"**.



Definitions

Term
Definition

Bit
A binary digit with two possible values, 0 or 1

Byte
A sequence of eight bits and has 256 possible

values from 00000000 through 11111111

kilobyte (kB)
1,024 bytes or 2^{10} bytes

Megabyte (MB)
1,048, 576 bytes or 2^{20} bytes

Gigabyte (GB)
1,073,741,824 bytes or 2^{30} bytes

Terabyte (TB)
1,099,511,627,776 bytes or 2^{40} bytes

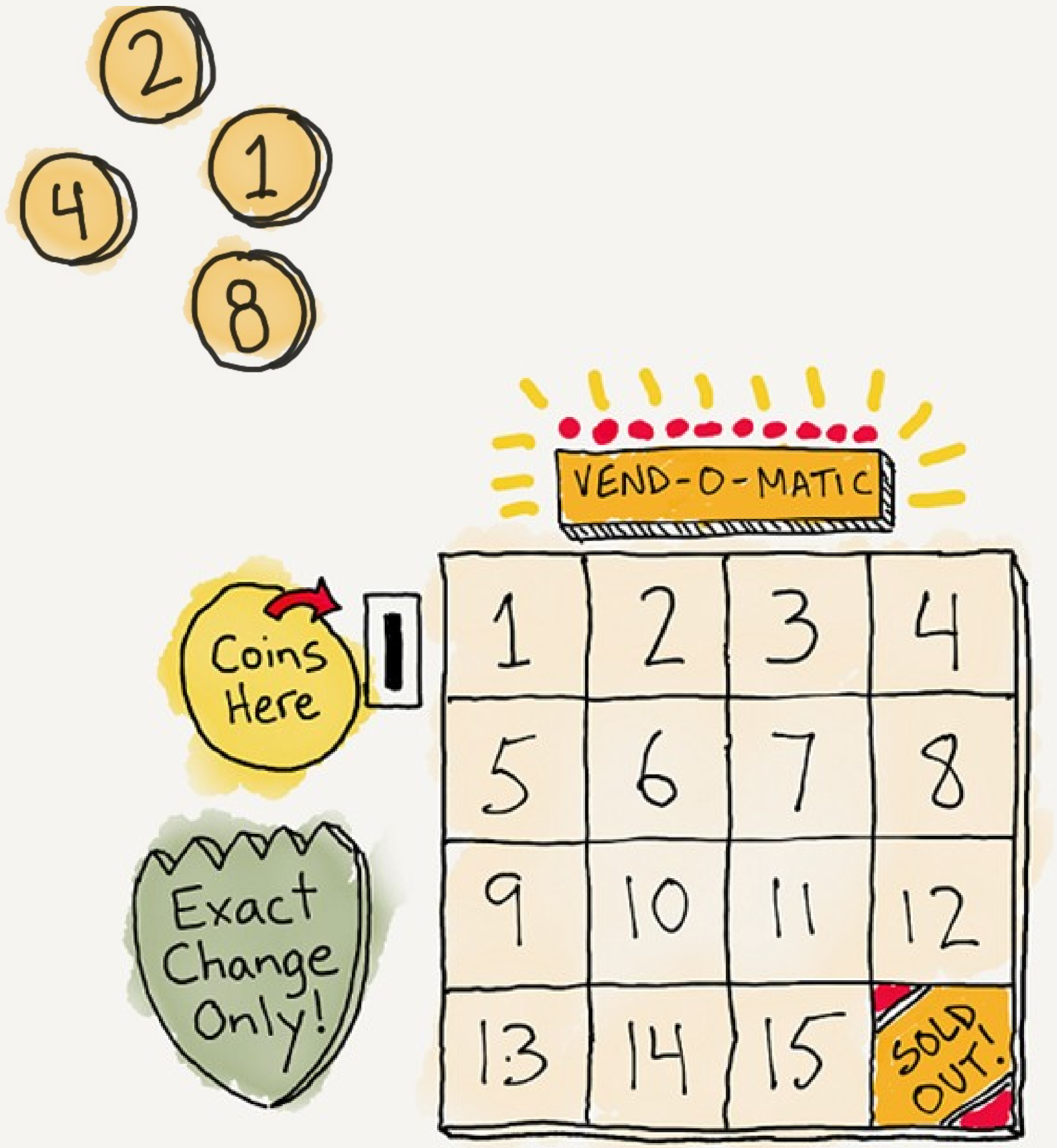
Binary vending machine

There is a vending machine that sells items at all prices.

- The machine cannot give change
- You must pay in exact amounts
- You have one of each coin: 1, 2, 4, 8

In pairs, create your own vending machine selling different items that cost 1-15 units.

Try buying different items from the vending machine.



Knowledge check

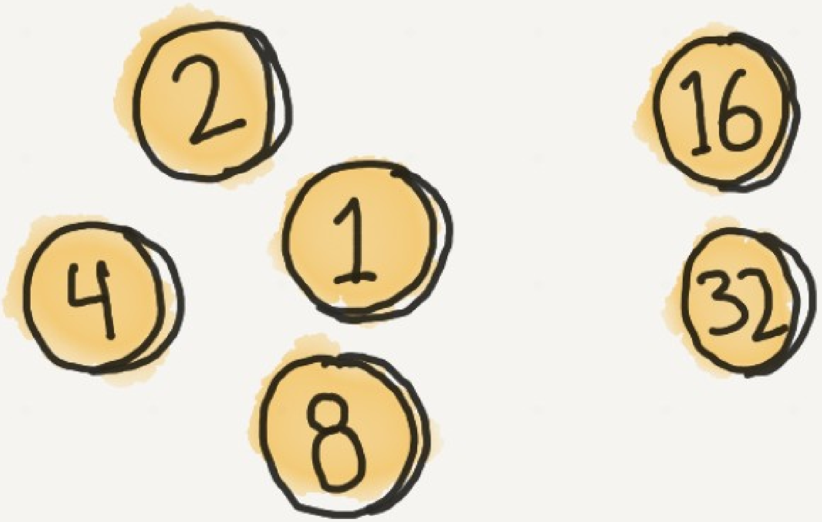
1. What is the price of the least expensive item you can buy?
2. What is the price of the most expensive item you can buy?
3. What else can you buy? What coin(s) would you use to do this?
4. What is the price of something you cannot buy because you don't have exact change?

Knowledge check (#5–6)

5. If we added two more coins to your set of coins, what is the next logical coin denominations? Why?

5. What is the new maximum price you could pay for an item?

These coins use the binary number system: each place value is 2 times the previous one.



Each coin is double the previous coin value

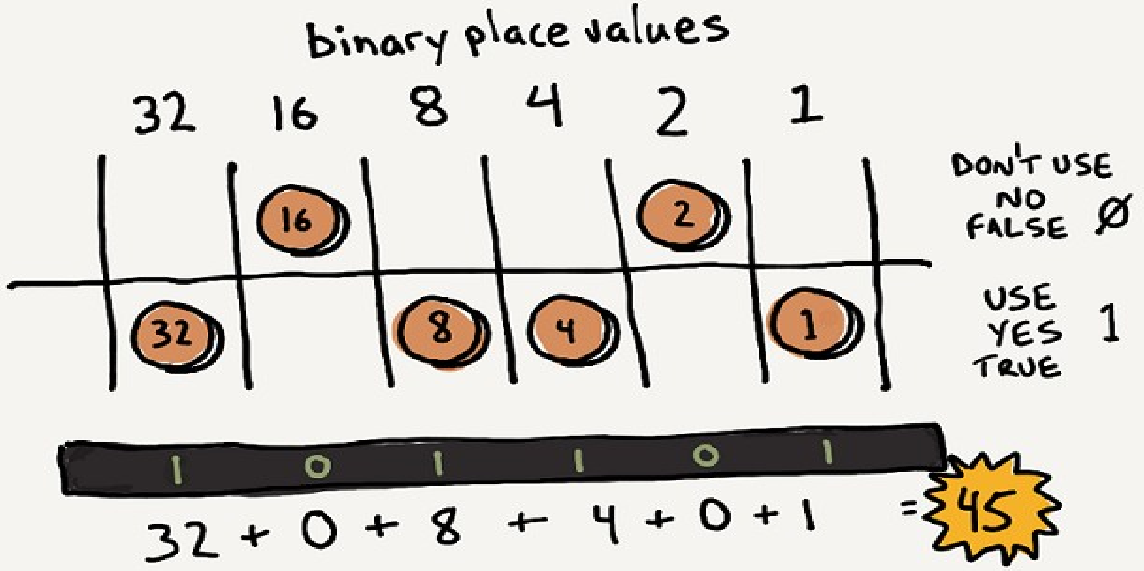
Knowledge check (#7)

Now, use 1s and 0s to represent the numbers instead.

Draw a table for your coins in descending order with 2 rows: one representing 0 and the other representing 1.

7. How would you represent the number 45 in this system?

   Try calculating different numbers in binary (base-2) notation



Binary:	1	0	1	1	0	1
Base 2:	32	16	8	4	2	1

32 + 0 + 8 + 4 + 0 + 1 = 45

What we learned in Lesson A

- Bits, bytes, and binary

Next time: Lesson B

- Code a binary transmogrifier
- Assess our learning with a quiz

Lesson B

Code a binary calculator

What we'll learn in Lesson B

- To code a binary calculator
- Review how much we've already learned

Activity: Binary calculator

In this activity, you will create a binary calculator with the micro:bit.

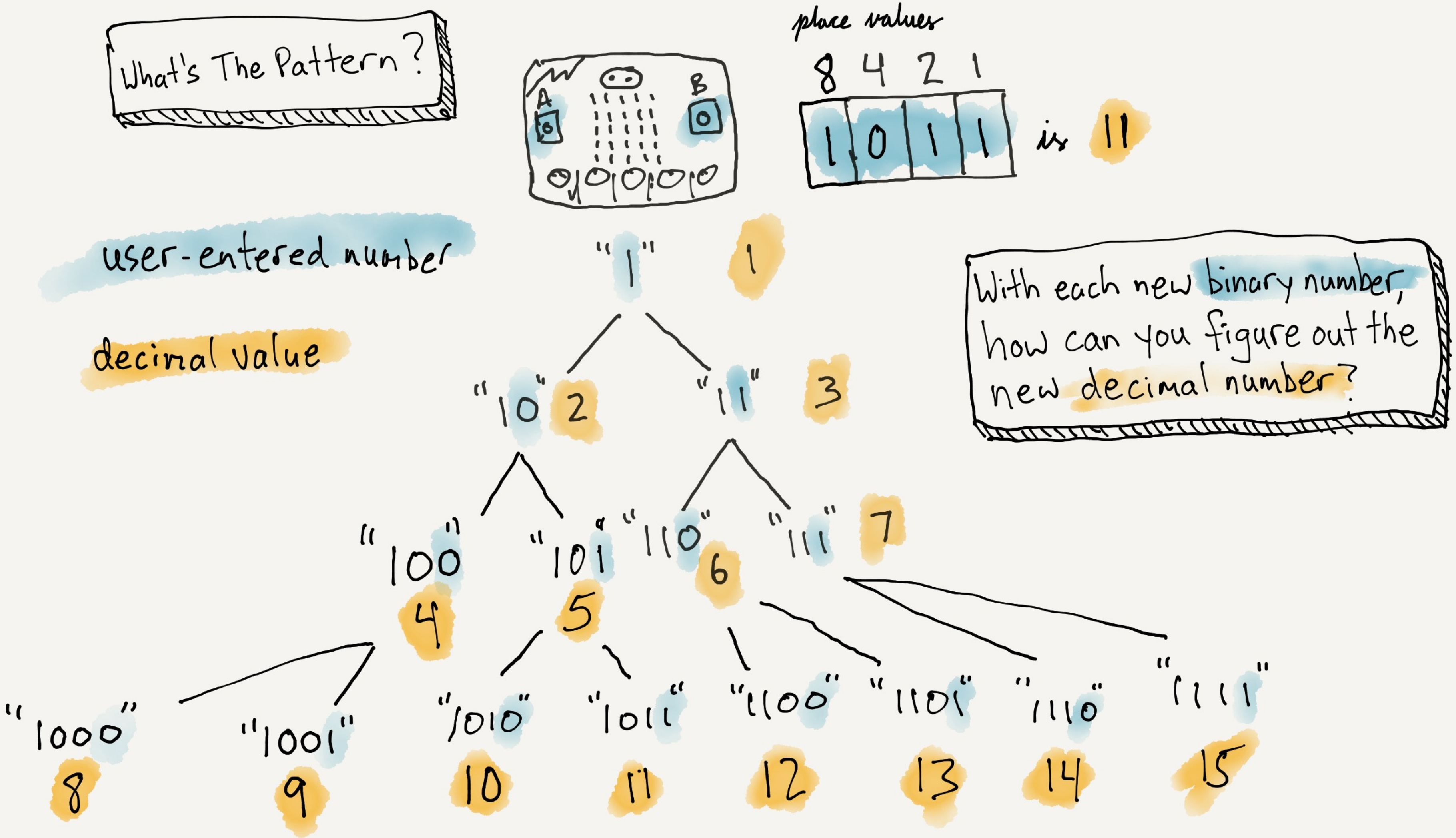
The user will be able to use the buttons to enter binary 0s and 1s and will be able to press **A+B** at any time to display the decimal equivalent of the number that has been entered.

The algorithm

- First, let's think about the Algorithm we need to convert a binary number into a decimal number.
- What's the pattern?
- When you add a 0 to the end of a binary number, how does the decimal value change?
- When you add a 1 to the end of the binary number, how does the decimal number change?

Binary	Decimal
1	1
10	2
11	3
100	4
101	5
110	6
111	7
1000	8
1001	9
1010	10
1011	11
1100	12
1101	13
1110	14

How does a binary calculator work?

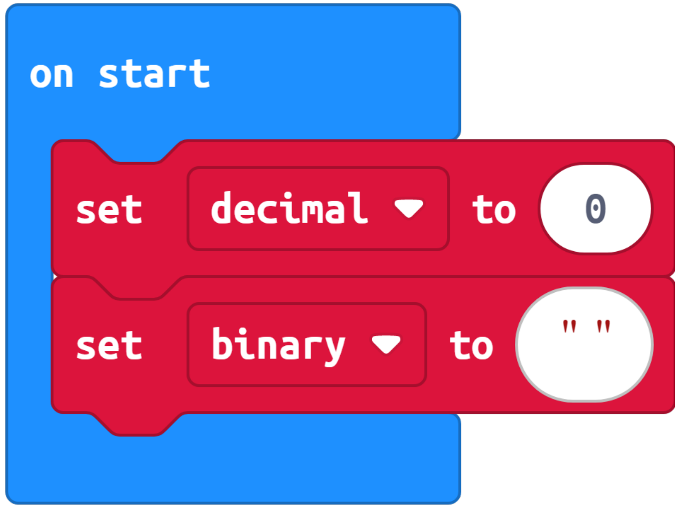


Coding a binary calculator

- Write out pseudocode for your program first, then in MakeCode.
- What happens when 0 or 1 is added to the end of a binary number?

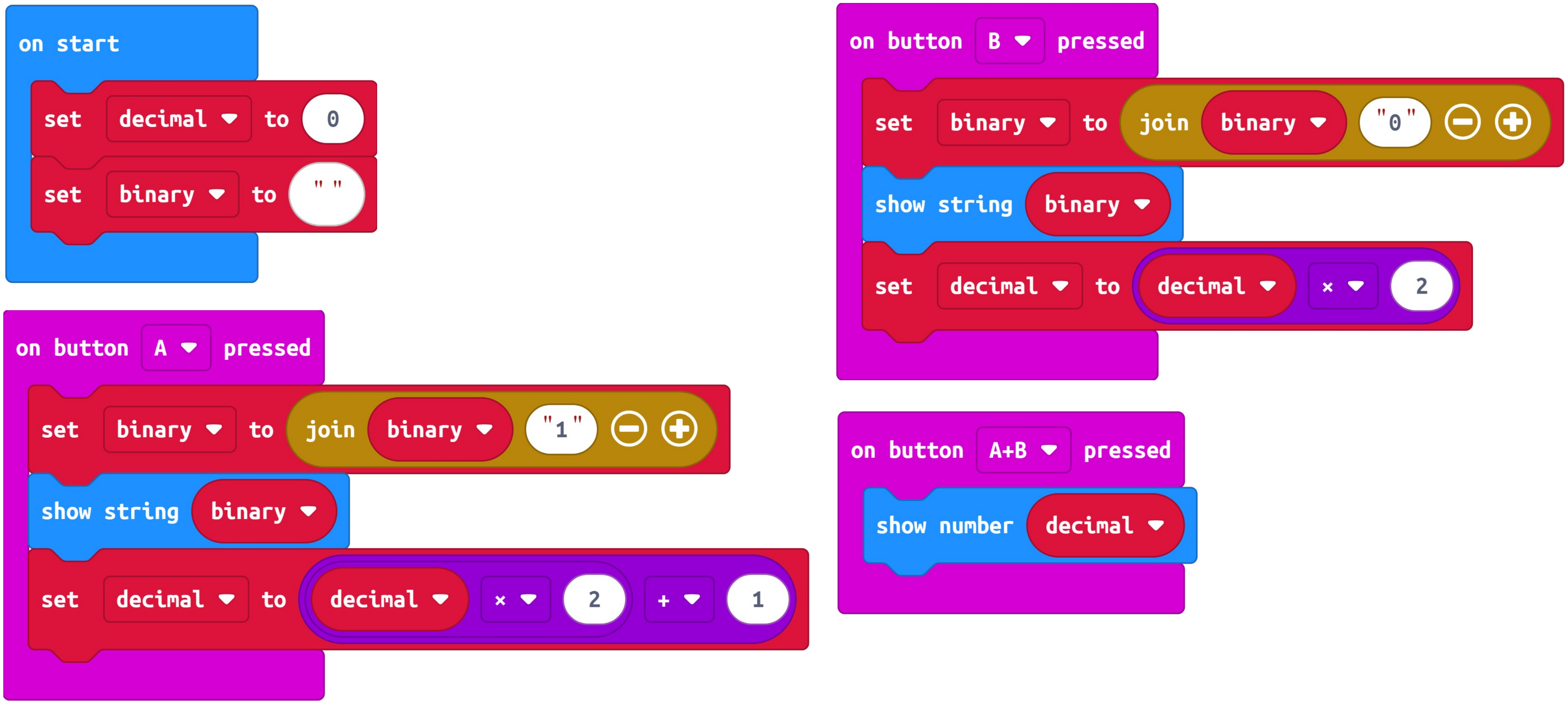
Hints

- Create variables for the Binary and Decimal values.
- For the Binary variable, use a String type to hold the sequence of 1's and 0's.
- Button A pressed – add a 1 to the end of the binary string
- Button B pressed – add a 0 to the end of the binary string



- Whenever someone enters a 0, the new decimal value is twice the previous value.
- If someone enters a 1, the new decimal value is twice the previous value, plus 1.

Your program might look like…



Mods

- Add a way to clear the binary and decimal values so you can start over.
- Add a way to erase the previous value.
- Create a decimal-binary converter that allows you enter a decimal value and see the binary equivalent when you press A+B.
- Create a physical housing or case for your Binary Calculator!

Time for a quiz!

What we learned in Lesson B

- Coded a binary calculator
- How much we've already learned

Next time: Lesson C

- Make a binary cash register

Lesson C

Make a binary cash register

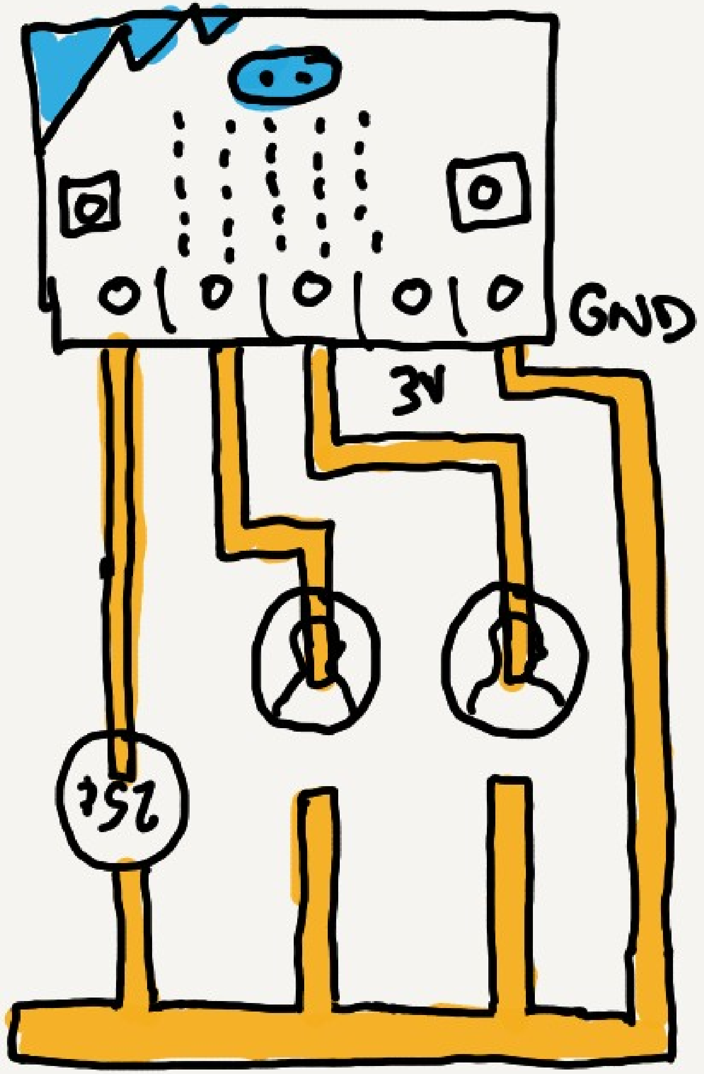
What we'll learn in Lesson C

- Make a binary counter and code it to display the decimal values of the numbers counted

Microsoft

Binary cash register

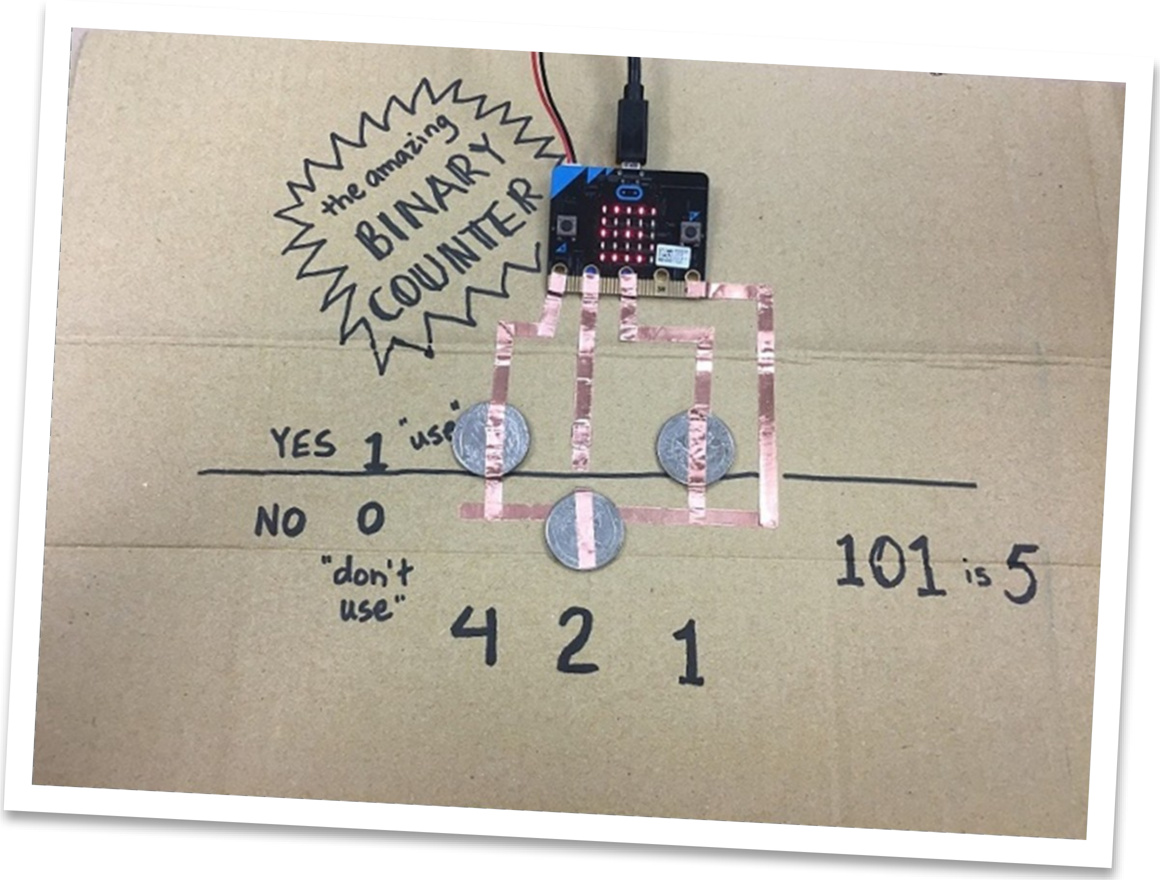
- The unplugged activity in Lesson A used a vending machine as a model for creating different combinations of binary place values.
- We found that for n coins, there is one and **only one way** to make every number between **0 and 2n-1**.
- Invent a paper and cardboard version of the binary counter, then program it to display the decimal value of those numbers.



Project example

This design uses coins and copper tape on a piece of cardboard.

- To indicate “off” or 0, the coins are flipped down.
- To indicate “on” or 1, the coin is flipped up so it lays flat across both pieces of copper tape, completing the circuit so:
  - The micro:bit can detect that the connected pin has been activated
  - It calculates and displays the decimal value of the binary number that is indicated by the coins



Project expectations

Use a design thinking approach and ensure the project meets these specifications:

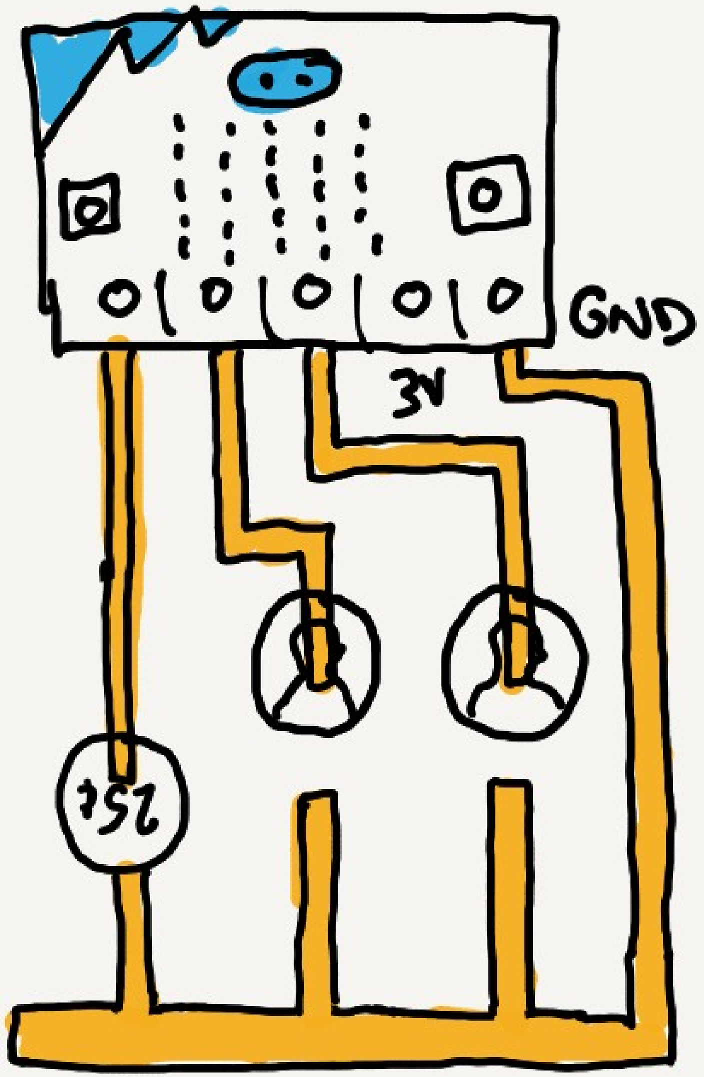
- Uses binary in a way that’s integral to the program
- Uses mathematical operations to convert decimal to binary
- All binary numbers display correctly
- The program compiles and runs as intended and includes meaningful comments in the code
- Provide the written Reflection Diary entry

Hints

- The micro:bit only has three pins, so your binary register is limited to three place values.
- You will need to use a strip of copper tape connected to the GND pin for the other side of the circuit for all 3 pins.

Mods

- Write some code that will display the number in binary when you press the A button.
- Think of a way to create more place values, perhaps by using a second micro:bit and a Radio connection.



Reflection Diary

- Describe what the physical component of your micro:bit project was (e.g., an armband, a cardboard mount, a holder, etc.)
- How well did your prototype work? What were you happy with? What would you change?
- What was something surprising to you about the process of creating this project?
- Describe one way in which your project differed from the example that was given. How would you recognize it as your own?
- Publish your MakeCode program and include the link.

What we learned in Lesson C

- Made a binary cash register